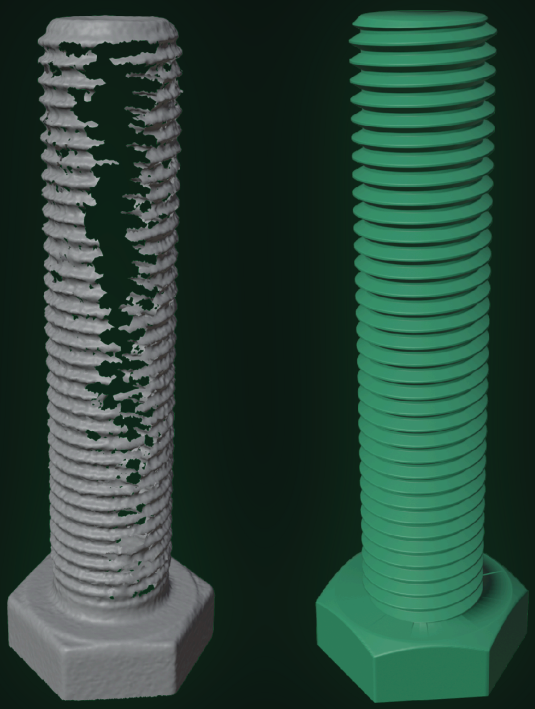
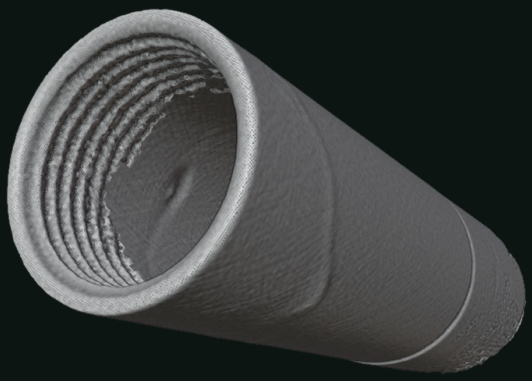
Scanned thread in. Clean thread out.

Rebuild the threads of a 3D scan as clean metric, inch or pipe threads: pins and holes, outside and inside, with a fit for 3D printing. Let it **find them all automatically**, or **click and paint** the ones you want. Every thread goes into a copy, so your scan is never touched.

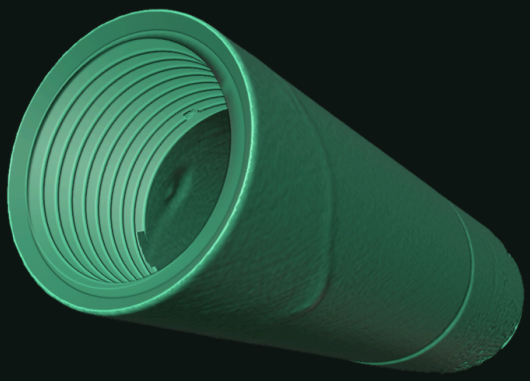
AUTO + MANUAL · EXTERNAL + INTERNAL · METRIC · INCH · PIPE · PRINT FIT



INTERNAL G1/2 · RAW SCAN



INTERNAL · REBUILT



Auto

one button finds every thread in a size range, pins and holes

Manual

click or paint to mark, then one Find Threads for all marks

In + out

male bolts and pipe ends, female holes and nuts, in one run

Safe

works on a copy; Esc cancels; a failed build is rolled back

01 GET STARTED

Install it. Select the scan. Press one button.

Install

Edit › Preferences › Add-ons › Install from Disk › **threadpharmacy.zip**, enable it, then **N › ThreadPharmacy** in the 3D Viewport.

Requirements

Windows (developed and tested on Windows, Blender 5.1 and 5.2 LTS; Linux and macOS are not tested). Blender 4.2 LTS or newer. No extra Python packages.

Large scans

16 GB RAM recommended for 10M triangles. Clean the scan first with MeshPharmacy.

License

GPL-3.0-or-later, full source. © 2026 Mirosław Pałucki.

FASTEST PATH · AUTO

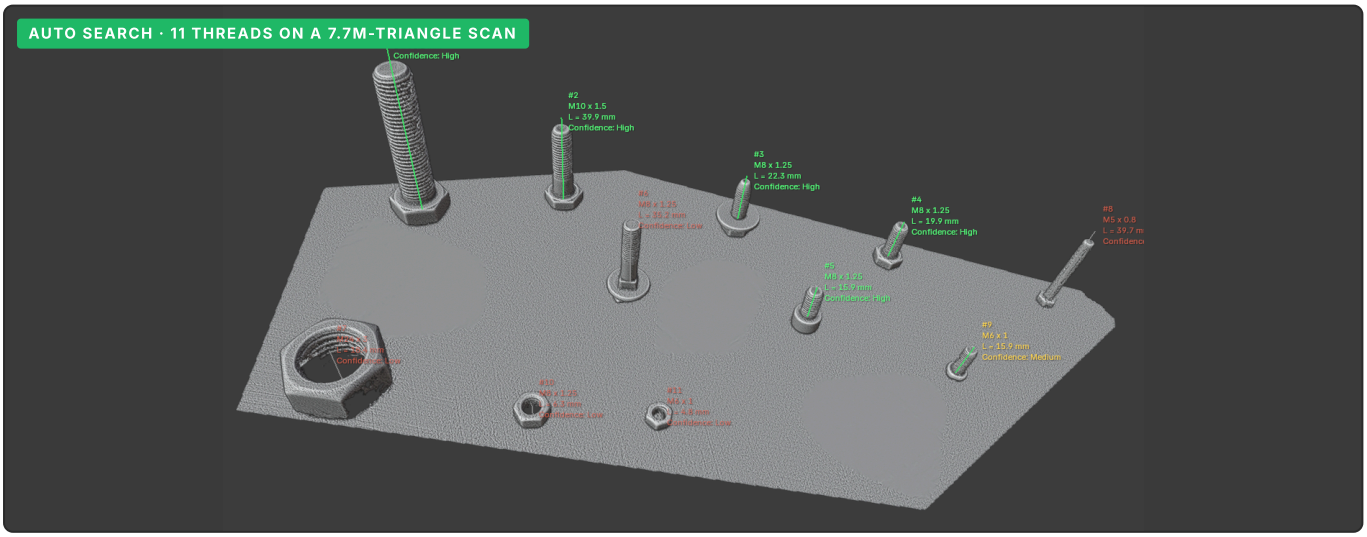
1. Select the scan, open **N › ThreadPharmacy › Auto search**.
2. Set the diameter range and **Both**; press **Find Threads**.
3. Check the list; High and Medium are ticked for you. **Change the thread type or the length** of any item before you build.
4. Choose the print fit; press **Build Checked**.
5. Look at the preview or the `name_threads` copy; **Export Report** if you need the numbers.

FULL CONTROL · MANUAL

1. Select the scan, open **Manual selection**.
2. Mark the threads: **Pick Thread** (click) or **Paint Area** (drag), in any mix.
3. Press **Find Threads** (or `Space`): all marks are analysed at once, up to 12 by default.
4. Check the list; set another **thread type** or **Thread length** for any item if needed.
5. Choose the print fit; **Build Checked**.

What is new in 1.0 and 1.0.4

- **Auto search and manual selection** as two tabs, with a shared found list.
- **Independent threads** on both ends of a pipe: male and female in one run.
- **Length detection**: a thread no longer runs along the whole part; a **Thread length** slider fixes the rest.
- **Very low** confidence and a “continue anyway?” question for a sliver of thread.
- **1.0.4 · Units mm | in**: every length can be entered and shown in inches (values are built in millimetres, pitch also as TPI).
- **1.0.4 · Long bolts**: the length now follows the thread up to 120 mm, so the top of a long bolt is rebuilt too.
- **One manual workflow**: Pick and Paint only mark (pink); **Find Threads** analyses every mark at once and turns them green.
- **Feedback**: a puff of magic dust on every mark, “analysing, please wait” and a wait cursor while it works.
- **Max threads** default 12 (up to 30) and **Max thread length** 100 mm, so a long edge is not taken for a thread.
- **Esc cancels** a search or a build, with *operation cancelled* and a restored scene.
- **Failed builds roll back**: no half-cut working copy.
- **Rebuild hex head / nut** as a tick box; print presets for Resin, PLA, PETG and ABS.
- **1.0.4 · Clean up after Build**: three tick boxes (small debris, small holes, big loose pieces) run after every build.



Mesh_Threaad_TEST_A, 7,681,399 triangles, range 4–34 mm, default limit of 12 threads. **5 High** (M16 × 2, M10 × 1.5, 3 × M8 × 1.25) · **1 Medium** (M6 × 1) · **5 Low**, left unticked: the M24 nut, the thin M5, two small nuts and one M8 bolt. No false thread: hits longer than 100 mm are skipped. Labels show size, length and confidence. Picture from an earlier run; version 1.0.4 gives **6 High**, **3 Medium** and **2 Low** on the same scan.

02 TWO WAYS TO WORK

Automatic when you can. Manual when you want control.

Both ways end in the same place: a measured thread, a size and a build into one working copy of your scan. Choose the tab at the top of the panel and switch whenever you like.

AUTO SEARCH

Find Threads

Searches the whole scan, or only a painted area. You set the limits, so a crowded scan never keeps you waiting.

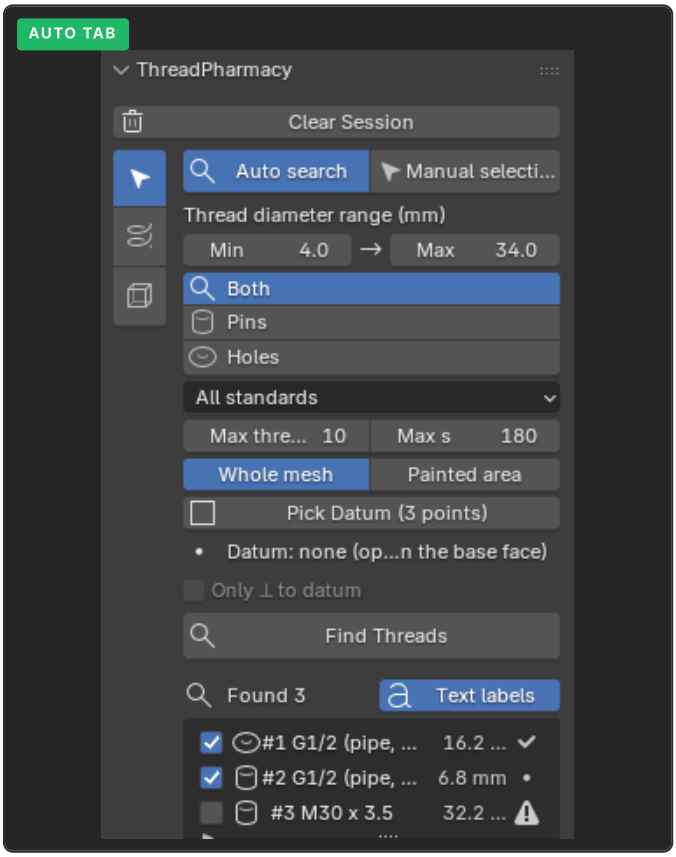
1. Set a **diameter range** (default 4–24 mm), **Pins**, **Holes** or **Both**, and the **standards** (All, Metric, Inch, Pipe).
2. Limit the **count** and the **time**. **Whole mesh** or **Painted area**.
3. Press **Find Threads**. A progress log runs; `Esc` stops it and restores the previous list.
4. Review the list: size, length, **confidence**. Untick what you do not want, then **Build Checked**.

MANUAL SELECTION

Pick Thread · Paint Area

For one thread, a damaged area, or a part with many near-identical threads. The tools stay active until you press Esc.

1. **Pick Thread**: one click per thread marks it pink. The ring shows the area; `Ctrl` +wheel resizes it.
2. **Paint Area**: drag to paint, `Ctrl` +drag erases. Clicks and paint can be mixed; a thread painted at each end of a pipe gets its own result.
3. Choose **Auto**, **Pin / bolt** or **Hole / nut**, the search radius and the standard. Optional: **Pick Datum** (3 points) for an exact axis.
4. Press **Find Threads** (or `Space`): every mark is analysed at once (up to **Max threads**, default 12) and turns green. Then change the thread type or length of any item before building.



Auto tab. Size range, Pins / Holes / Both, standards, count and time limits, Find Threads, then the found list with a confidence icon per thread and All / None / High only.

MANUAL · MARK, THEN FIND

1. **Mark** with Pick Thread (click) or Paint Area (drag). Marks are pink.
2. Mark as many threads as you like, pins and holes in one go.
3. Press **Find Threads** or `Space`. A puff of magic dust, then "analysing, please wait".
4. Each click or separate painted area becomes one entry in the list. The surface turns green.
5. For any entry: change the **thread type** (another size, pitch or standard) and the **length** before you build.

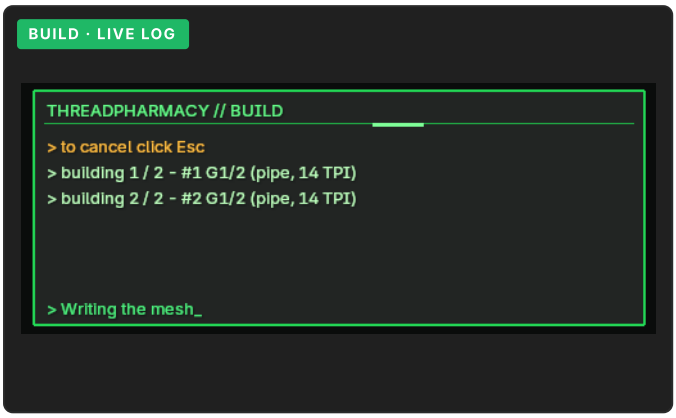
Same list, same Build as the Auto tab. `X` or **Clear Marks** removes marks that were not analysed yet.

Manual workflow. One consistent flow for click and paint, with the same default of 12 threads as Auto.

03 BUILD · SAFETY · CONFIDENCE

It tells you what it is sure of, and never touches your scan.

Press **Build Checked**. The old thread is cut out along the measured axis, the clean ISO or pipe thread is built in, and the rest of the scan stays exactly as scanned. A live log shows the progress, and you can stop at any time.



Two threads, 68.6 s on the 894k-triangle pipe scan (exact Booleans, local patches). The log says how to cancel: `Esc` ends the run and the scene is restored.

Always a copy

The first build duplicates the scan as `name_threads` and hides the original. Every further thread goes into the same copy. Ctrl+Z works as usual.

Cancel any time

Find Threads and Build show *to cancel click Esc*. A cancelled search puts the previous list back; a cancelled build undoes everything that run did, and says *operation cancelled*.

Failures are rolled back

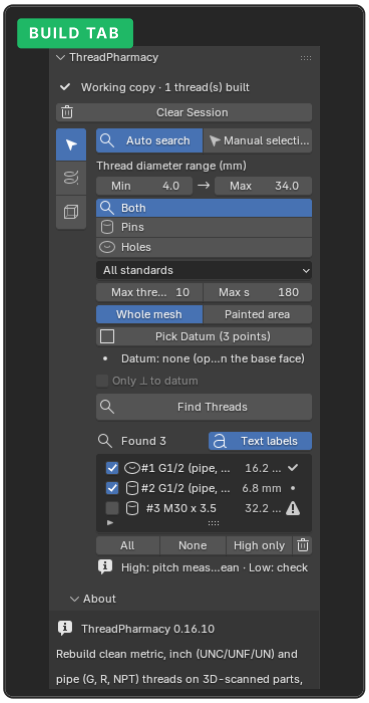
If one thread cannot be built, its changes are undone and the working copy stays whole; the other threads are kept.

Confidence for every thread

LEVEL	MEANING	TICKED
High	Pitch measured, thread seen all round.	yes
Medium	Clean surface, size from the diameter.	yes
Low	Check the size, or pick it from the list.	no
Very low	Too little thread seen: a sliver or a blob.	no

Painted only a sliver?

ThreadPharmacy checks how much of the thread was seen: coverage around the axis, number of points and length against the pitch. If it is not enough, the entry is marked **Very low**, stays unticked, and building it asks *"Not enough data to determine the thread. Continue anyway?"* before anything is changed.

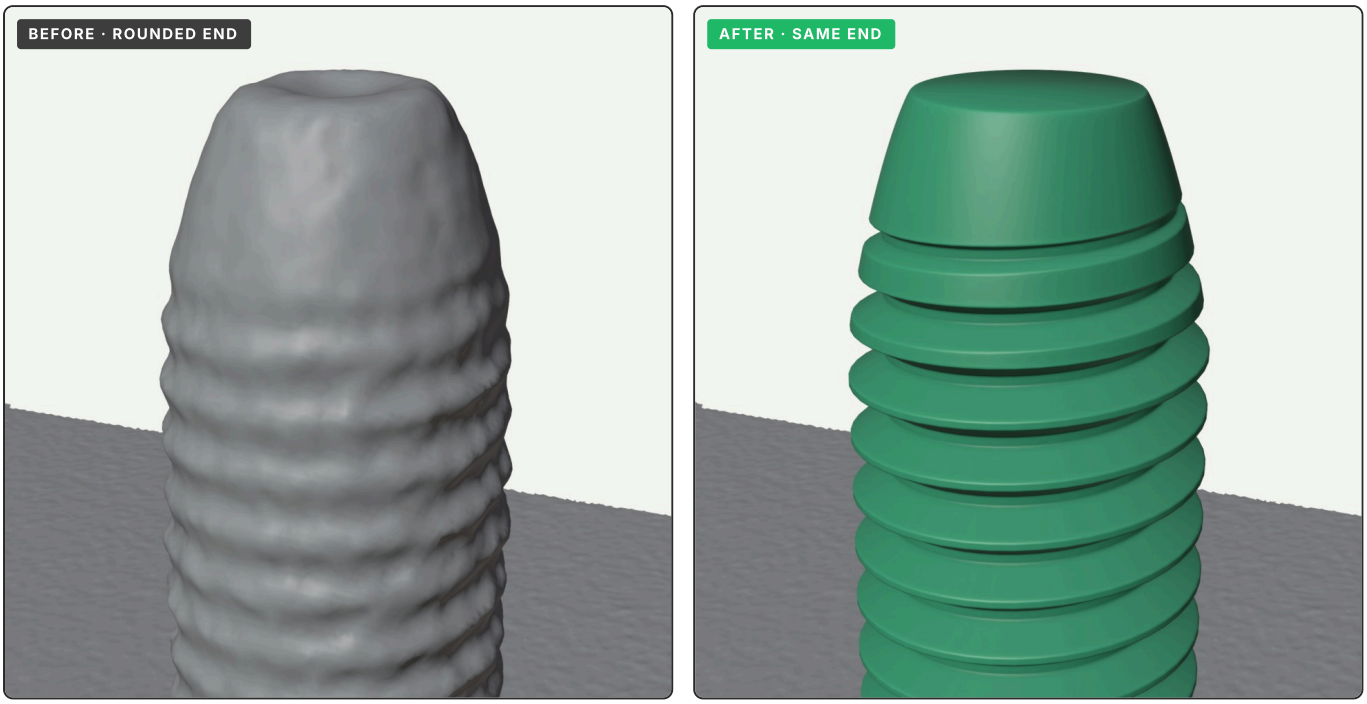


Build tab: print fit, **Rebuild hex head / nut** tick box, quality, cut margin, Boolean solver, Build Checked, Preview, clean-up, Export Report.

04 MEASURE & PRINT FIT

Measured on the scan. Made to fit when printed.

For the right pitch every point of a thread collapses onto one tooth profile. ThreadPharmacy tries every candidate pitch and keeps the one that stands out, then builds the exact profile offset by the clearance you choose.



M8 × 1.25 bolt with a rounded (cone) end, from a blue-laser scan. Length and curve are measured, and the new pin gets the same end. Chamfered and dog-point ends are recognised too.

Print fit presets

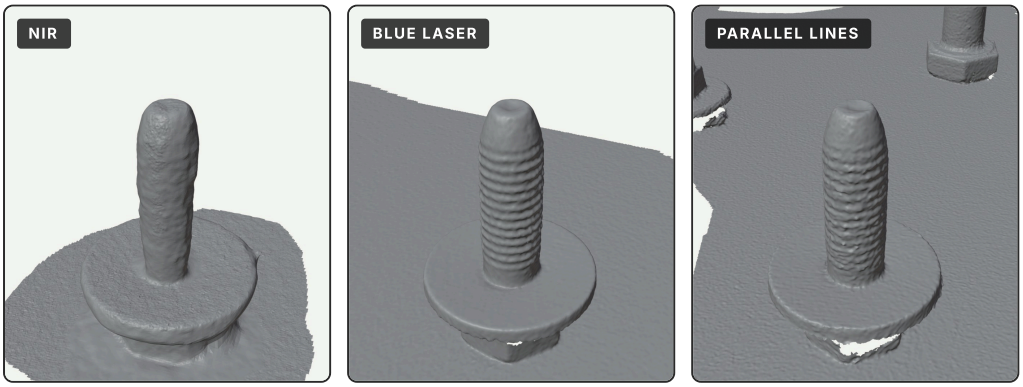
PRESET	RADIAL	USE
ISO nominal	0 mm	Exact basic profile, for CAD or machining.
Resin / SLA	0.10 mm	Tuned resin printer.
PLA	0.20 mm	Tuned 0.4 mm nozzle.
PETG	0.25 mm	PETG oozes: a little more.
ABS	0.30 mm	ABS shrinks while cooling.
Custom	any	Set the clearance yourself.

Pin smaller, hole larger, all round. Rules of thumb, not standards: print a test nut first.

What is measured

- **Axis** from a cylinder fit on oriented points; floors, heads and hex flats are ignored.
- **Base face** found along the axis, or three datum points you pick.
- **Pitch and hand** by helical unwrapping; accepted only when it clearly beats every other candidate.
- **Size** on the pitch diameter, which works whether the scanner resolved the teeth or smoothed them.
- **Length** from the scan, adjustable with a slider.
- **Taper** measured: it separates G from R and NPT.

Same bolt, three scanners · M8 bolt, L 22 mm · default settings, earlier benchmark

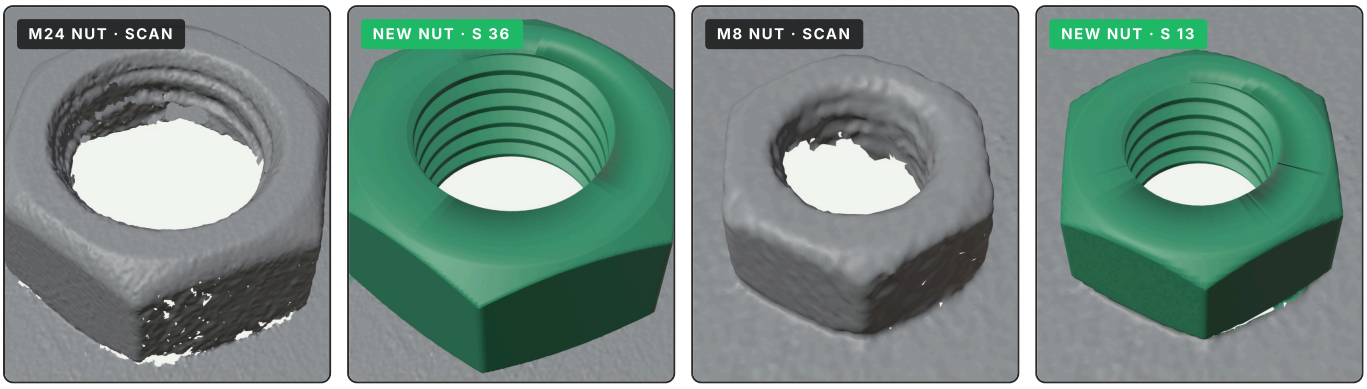


SCANNER	RESULT
NIR	M8 × 1.25 · Low
Blue laser	M8 × 1.25 · High
Parallel	M8 × 1.25 · High

05 NUTS, STANDARDS & BIG SCANS

Bolts, nuts, pipes and fittings. Scans with ten million triangles.

For a hole the scanned wall is filled and the new thread is cut back in. Through holes get a through thread, blind holes get a 118° drill point. Tick **Rebuild hex head / nut** and a recognised hexagon is rebuilt at the ISO / DIN width across flats.



Inside a nut the scanner sees little, so the size comes from the diameter (Low confidence): check it, or pick the size from the list. The hexagon is found from its six flats (M24: 35.0 mm measured → 36 mm ISO) and rebuilt with 30° chamfers.

Thread standards

FAMILY	STANDARDS	SIZES
Metric	ISO 261 / 68-1 coarse and fine · DIN 2353 hydraulic	M1 – M64
Inch	UNC / UNF (ASME B1.1) · UN-12/16 for JIC, ORB, ORFS (SAE)	#4 – 1-1/2"
Pipe, parallel	G / BSPP (ISO 228-1), 55° Whitworth	G1/16 – G2
Pipe, taper 1:16	R / Rc (BSPT, ISO 7-1) · NPT (ASME B1.20.1)	1/16" – 2"

Limit the search to what your parts can be: **All, Metric, Inch, Pipe** or Metric + pipe. The pitch separates look-alikes (M8 × 1.25 / 5/16"-18), the measured taper separates G from R and NPT.

Why big scans stay fast

- Only the part around the thread is cut out, processed and welded back by exact vertex position.
- Removal bodies are split into small facets, which keeps the Exact solver quick.
- Booleans: Manifold (Blender 4.5+) first, Exact as fallback, every result checked and run in a helper process.
- Search runs in slices with a progress log; the Blender window never says “Not responding”.

Measured on real scans

SCAN	TRIANGLES	FIND	BUILD	NOTE
Pipe, G1/2 pin + hole, M30 bolt	894k	3 found · ~1 min	2 threads · 68.6 s	M30 scan broken → Low, unticked
NIR, 2 bolts	409k	2 of 2	~10 s	threads smoothed by the scanner
Blue laser, 8 bolts + 3 nuts	6.1M	11 of 11 · 100 s	~12 s per thread · 2.9 GB	6 High, 1 Medium, 4 Low
Parallel lines, set of 11	10.0M	11 of 11 · 166 s	18–28 s · 4.9 GB	scans stay open, no new open edges

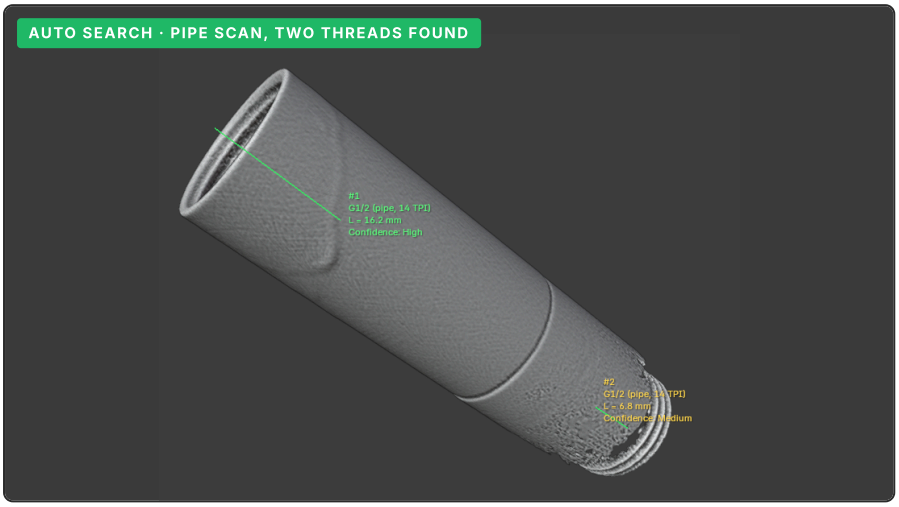
Scans that are open (they end at the table) stay open. Sizes of these benchmarks come from the same scans and defaults; the pipe row was measured with the current version.

Limits. Smooth scans (NIR) and small threads: size from the diameter only, so check it. Check inch sizes and large taper holes (NPT / Rc 3/4" and up). Deep, narrow holes are often only partly scanned. A search radius that is too small (a few mm) gives nonsense. Fittings: thread only, no cone or O-ring seat. Very ragged ends can leave scraps of the old rim beside the new thread.

06 A ROUGH SCAN IS FINE TOO

One pipe, a poor point cloud: both threads still found.

This pipe is a rough scan: noisy, torn at the end and with only a few millimetres of thread on the male side. ThreadPharmacy still found the internal G1/2 in the hole and the external G1/2 on the pipe end on its own, and rebuilt both, as the real before and after below show. Each thread has its own axis, length and result even though they share one line.



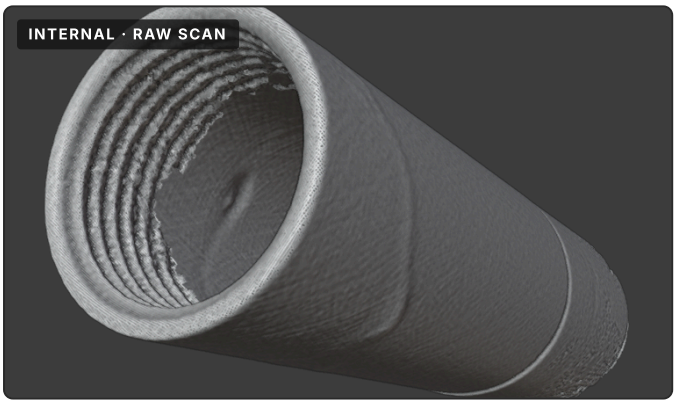
Found in about a minute

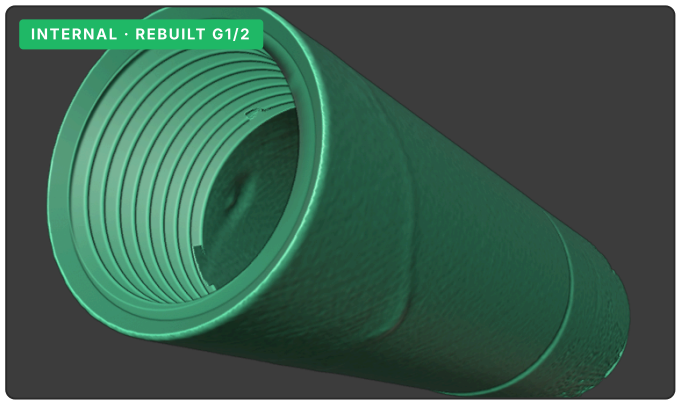
553k-triangle pipe scan, default settings with the range widened to 34 mm.

#1 G1/2 hole
L 16.2 mm · High

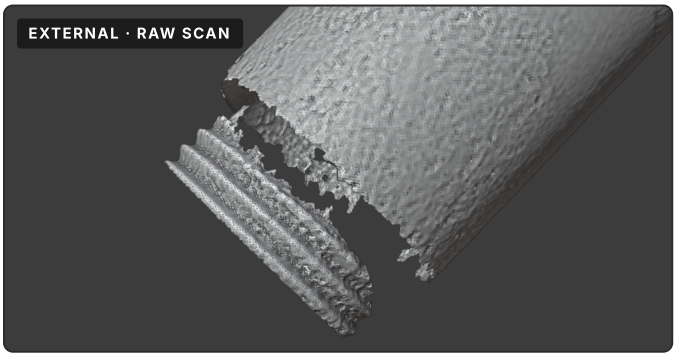
#2 G1/2 pin
L 6.8 mm · Medium

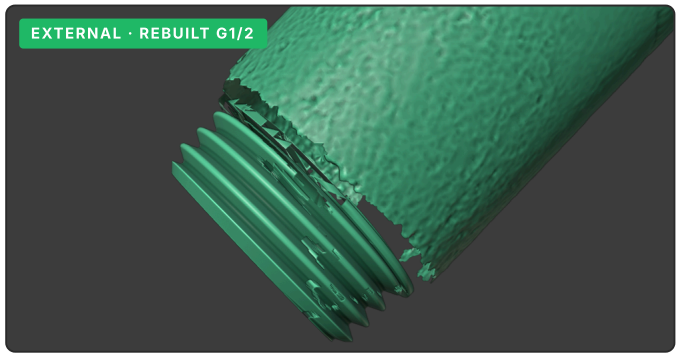
Each end is its own thread, with its own length.





Hole. The torn, smoothed scan of the thread becomes a clean G1/2 (14 TPI) internal thread, with the old surface filled and the new thread cut back in.





Pipe end. Only about nine millimetres of thread, damaged in the scan, and the length is measured from the scan instead of threading the whole pipe. Where a scan is ragged, scraps of the old rim can remain beside the new thread; the tick boxes under **Clean up after Build** (small debris and small holes on by default, big loose pieces optional) run after every build, and the same three steps can be run by button on the finished result. The result is yours to check.

Pins and holes

Pin or hole is read from the direction of the surface normals. Bolts, studs, pipe ends, nuts and tapped holes.

Length is measured

A thread ends where the surface steps up to a body or collar. Wrong? Use the **Thread length** slider, or switch auto length off.

Independent results

Paint or pick each end separately. Two threads on one axis never merge, and a failed second build leaves the first one intact.

07 TIPS & TROUBLESHOOTING

When a result is not what you expect: what to try, in this order.

Notes from testing on bolts, nuts and pipes of very different scan quality, from blue-laser and NIR scanners to a rough point cloud. Most problems come from the scan, not from the thread. A good first step is to check the **confidence** of the item, because it tells you how much to trust the size.

SYMPTOM	LIKELY CAUSE	WHAT TO TRY
A thread is not in the list	Outside the diameter range, search seed missed it, or a thread of the same size already took the slot.	Widen the **diameter range** and raise **Max threads**; or switch to **Manual selection** and **Pick Thread** in the middle of the thread, or **Paint Area** over it.
Nut or hole not found	Only the face of the nut was scanned, not the bore wall. There is no thread surface to measure.	Scan the inside (tilt the part, use a smaller scanner stand-off). The hex outside alone can confirm a size but not build a thread.
Thread stops short of the real end	Length is measured from connected thread surface; a gap in the scan, or a wide step, ends it early.	Raise **Thread length** in the item (or switch auto length off) and set **Max thread length** high enough. Long bolts above 100 mm need the limit raised.
Axis looks slightly tilted	Short thread, noisy scan or Low confidence: the axis is fitted from the thread alone.	Use **Manual selection**, **Pick Datum** (3 points on the flat base face) then **Find Threads**. The axis becomes exactly square to the datum. Check the size, because confidence is reported as Low in this mode.
Wrong size or pitch suggested	Smoothed teeth, or the diameter lies between two standards.	Open the item and pick the **thread type** from the list; for inch parts choose UNC / UNF explicitly. Try a smaller **search radius** so only the thread is analysed.
Result is not watertight	The scan itself is open, or a nut is built from separate sheets.	Close holes first (MeshPharmacy or **Fill Holes**), then rebuild. A source that is not closed cannot give a closed result.
Hex head or nut not rebuilt	At least some of the side flats must be visible to measure the hex.	Tick **Rebuild hex head / nut** only when flats are in the scan; otherwise leave the head as scanned.

Parameter combinations that worked

- **Clean scan, many parts:** Auto search, range = smallest to largest size you expect, **Both**, default limits, then untick Low items you do not need.
- **One awkward thread:** Manual, **Pick Thread** in the middle, search radius just above the thread radius, then fix the thread type by hand.
- **Square to a flat plate:** Manual, **Pick Datum** on the plate, then Pick Thread. Best for perpendicular bolts and studs.
- **Ragged or partial scan: Paint Area** over the best part of the thread; a sliver is marked **Very low** and asks before building.
- **3D print fit:** PLA 0.20 mm, PETG 0.25 mm, resin 0.10 mm; print a short test nut or test bolt before the full part.

Good habits

- **Prepare the scan:** remove noise and islands, and decimate extremely dense meshes, before searching. 10M triangles work, but need about 16 GB RAM.
- **Do not work while it works:** a build on millions of triangles runs in the background; wait for the log to finish or press `Esc`.
- **Check the copy, not the original:** the result is in **name_threads**; your scan stays hidden and untouched. `Ctrl+Z` undoes a build.
- **Check each thread:** compare the preview with the scan, and use **Export Report** if you need the numbers.
- **Re-run a single item:** change its type or length in the list, untick the rest, **Build Checked**.

What the datum plane does

You pick 3 points on a flat face next to the thread (the plate a bolt stands on). The axis is then made exactly **square to that plane** (tick **Axis ⊥ datum**, on by default), the thread starts **at the plane** so the length is measured from it, the base face itself is **ignored** (0.4 mm band), and the report shows the **angle and the deviation in mm** between the measured axis and the datum. Without a datum the base face is found automatically and the axis is snapped only within 8 degrees (25 for a short thread). Use it when the axis comes out tilted, but not for a part that really is tilted: the axis is forced square. The **mm | in** switch at the top of the panel changes every length field and label.

Honest limits

A thread can only be rebuilt where the scan shows it. Where the surface is missing, smeared or very short, ThreadPharmacy marks the result **Low** or **Very low** instead of guessing. Treat every result as a good starting model that you check against the part, and a print fit as a starting point, not a standard.